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Chronic Suppression of a Multidrug-Resistant *Pseudomonas aeruginosa* in Prosthetic Joint Infection Using Personalized Bacteriophage Treatment

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Abstract

Multidrug resistant (MDR) bacterial infections without antibiotic options are a public health emergency. Infections associated with medical implants serve as an example. Conventional antibiotics have limited ability to eradicate these infections as they are associated with antibiotic-tolerant biofilms. Here, we report the use of bacteriophage therapy for the treatment of a MDR, non-operable *Pseudomonas aeruginosa* periprosthetic joint infection that had failed multiple antibiotic and surgical interventions. Treatment with intermittent bacteriophage therapy alone without antibiotics over a 2 year time period resulted in clinical resolution of the infection, but not microbiological eradication. Bacteriophage therapy established this control, in part, by altering virulence as defined by disease severity and symptoms and disrupting biofilm. Whole genome sequencing demonstrated the continued presence of bacteriophage during treatment. This provides preliminary evidence that bacteriophage therapy can be used to treat MDR infections in salvage cases when surgical and antibiotic options do not exist.

Introduction

Medical implants and devices have transformed medicine by providing a solution to a wide range of problems without a medical solution, transforming surgical disciplines including cardiothorasic, vascular, dental, neurological, and orthopaedics. About half of all nosocomial infections are associated with medical devices[1]. These implant-associated infections are difficult to treat because of their association with antibiotic tolerant biofilm[2].

An example of this challenge includes total knee and hip arthroplasty (joint replacements). It is one of the most common surgeries in the developed world, and the most severe complication includes periprosthetic joint infections (PJI)[3]. It is a devastating diagnosis, often requiring multiple surgical procedures over several years with a five-year mortality between 20 to 25%[4,5], higher than many cancers[5].

Periprosthetic joint infection is difficult to treat because of its association with antibiotic tolerant biofilms. Eradication of the infection requires the removal of components, but surgical intervention with implant removal can have significant morbidity or mortality. Periprosthetic joint infection is associated with multi-drug resistant (MDR) hospital-acquired ESKAPE organisms (*Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, Enterobacter spp,* and *Escherichia coli*). *P. aeruginosa* is a particularly challenging pathogen as conventional antibiotic treatments are often ineffective[6,7]. There is thus an urgent need to develop alternative therapies that can overcome the limitations of antimicrobial-based treatment strategies for implant-associated infections.

Lytic bacteriophages, commonly known as phages, have the potential to be an effective therapy against these difficult-to-treat infections because of their innate anti-biofilm activity[8,9]. Phage therapy has been used as an adjuvant to antibiotics to treat PJI and other biofilm-associated infections[10-12,13]. A common strategy includes combining long-term conventional antibiotics with bacteriophage therapy to eradicate or suppress the complex infection.

In this study, bacteriophage therapy was utilized in a unique way as a stand-alone therapeutic without antibiotics. A patient presented with a chronic multidrug-resistant *P. aeruginosa* PJI that was non-operative. They had been referred to palliative care after multiple surgical and antibiotic treatment failures. After standard of care treatment, the patient remained culture positive, but intermittent dosing of bacteriophage resulted in clinical control. This was achieved, in part, through long-term bacteriophage propagation in the patient, limiting biofilm formation, and a phenotypic decrease in the bacteria's ability to cause disease symptoms and severity. This example serves as a proof of concept supporting the need for further clinical studies.

Results

Clinical Course and Bacteriophage Therapy

A 96-year-old male with multiple comorbidities including heart failure presented with a chronic multidrug-resistant *Pseudomonas aeruginosa* infection in complex implants. A detailed clinical course and treatment protocol is provided (**Fig. 1a**, **Supplementary information, Supplementary Fig. 14**). Staged total knee and hip arthroplasties were performed more than 15 years previously that had been complicated by a periprosthetic fracture (**Fig. 1b-e**). Later, he was diagnosed with a *P. aeruginosa* PJI of his knee that metastasized to his ipsilateral hip (**Fig. 1f-g**). Initially sensitive to oral antibiotics, the organism became multidrug resistant following chronic suppression with ciprofloxacin. Chronic suppression was no longer feasible given the bacteria was resistant to all oral antibiotic options. There were multiple attempts to discontinue intravenous antibiotics over an approximate 2 year period, but this resulted in painful symptomatic infection or sepsis. During each of these recurrences and multiple surgical interventions, the cultures continued to be positive for MDR *Pseudomonas*.

The patient presented to our institution with a MDR *Pseudomonas* PJI and complex implants with limited treatment options. At the outside institution, intravenous antibiotics were planned to be stopped permanently with plans for palliative care and hospice. Traditional treatment options were limited including observation with discontinuation of antibiotics, repeat debridement and antibiotics with implant retention (DAIR), which had failed the previous four times, removal of components, or amputation. These options had a high probability of leaving him non-ambulatory and with a high expected mortality[5,14].

Given limited treatment options, FDA-expanded access was obtained to start bacteriophage therapy. Cultures were obtained and two bacteriophages (PASA16 and Φ83)[12,15] were identified with planktonic and biofilm activity against the *P. aeruginosa* clinical isolate tested. Titers of phage for dosing were selected based on maximizing phage titer below the threshold of endotoxin exposure (5 endotoxin units (EU)/kg/hour as determined by the United States Food and Drug Administration (FDA)[16], (approximately 4.7 x 10^{10} plaque-forming units, PFU/mL, PASA16 and 7.5 x 10^{9} PFU/mL Φ83). Standard of care treatment was provided with (DAIR) using intermittent bacteriophage therapy as adjuvant care. The initial dose was intra-articular followed by a daily intravenous dose for one week. The final treatment protocol is included as supplemental data (Bacteriophage dosing and treatment protocol).

Diagnostic markers indicated continued infection despite clinical resolution

The patient was transitioned to chronic suppression with intermittent bacteriophage treatment. Following standard of care treatment, the knee aspiration continued to meet the Musculoskeletal Infection Society (MSIS) and the International Consesnsus Meeting (ICM) definition of infection [17 18]. Plaque assay on the synovial fluid prior to cycle 2 phage administration demonstrated the continued presence and propagation of bacteriophage from cycle 1 for more than 2 months (**Supplementary Fig. 12)**. The patient remained symptomatic with persistent knee effusion and pain at the site for 3 months. Intraarticular bacteriophage dosing was therefore repeated, and the effusion and pain subsided. Based on the initial success and catastrophic consequences of uncontrolled infection, bacteriophage dosing was continued to be dosed at approximately every 1-6 months without antibiotics. Aspirations completed prior to dosing demonstrated that the patient remained culture positive (**Fig. 1a**) and met the International Consensus Meeting (ICM) 2018 definition of prosthetic joint infection[18] (**Fig. 2a-c**) but he continued to remain clinically asymptomatic. At 6 months, the knee effusion resolved. This corresponded with an overall decrease in inflammatory markers (**Fig. 2c, Supplementary Table 4**). No significant adverse events were observed while on phage therapy. There was a transient elevation in liver enzymes which other groups have previously reported[12] (**Fig. 2d**). The patient has remained off antibiotics and has remained asymptomatic at 2-year follow-up with baseline ambulatory function comparable to prior to infection. Long-term clinical control of the infection was established off of antibiotics for more than 2 years. Prior to use of bacteriophage therapy, discontinuation of antibiotics resulted in a painful leg with episodes of sepsis.

Phage lytic activity against patient isolates over time Given the extended time period of treatment, phage activity in pre- and post-phage bacterial isolates was examined for phage sensitivity. For analysis, we selected pre-phage (S1 and S2), mid-treatment (S8), and long-term treatment (S22 and S23) *pseudomonas* isolates. The phages used for treatment, PASA16 (Family: Lindbergviridae, genus: Pbunavirus*;* based on morphology: contractile tail is a main characteristic of myoviruses; genome size: 66,127 base pairs) and Φ83 is confidential however, based on morphological analysis of the TEM images, it is classified as a myovirus (**Fig. 3a, b**), were assessed for activity using standard growth inhibition and a plaque assay **(Fig. 3c, d; Supplementary Table 5)**.

For both PASA16 and Φ83, growth inhibition assays on pre-phage, mid-term, and long-term phage treatment isolates demonstrated no statistically significant loss of activity (**Fig. 3c, d**). There was a morphological difference between the pre-phage and long-term treatment isolates obtained from the patient. The long-term treatment isolates exhibited a small colony variant phenotype; however, they did not show different phage susceptibility (**Supplementary Fig. 1a, b**).

Neutralizing antibody production

Given the multiple doses, the patient's host immunological response to bacteriophage therapy was investigated. Neutralizing antibodies have been observed in serum following intravenous phage therapy in multiple case reports[19,20]. As treatment progressed, we began to monitor for phage-neutralizing antibody activity initially in the serum and then in synovial fluid. Patient specific neutralizing antibodies to these phages were monitored by assessing changes in phage lytic activity when performed in serum or synovial fluid. For treatments involving PASA16, serum collected at 6, 9, 12 and 24 months showed a significant high neutralizing activity and reduced the PFU count (P < 0.01; **Fig. 3e**) when compared to the SM+ buffer control. Incubation of the phage with patient synovial fluid showed a significant reduction (P < 0.007, **Fig. 3f**) in PFU count when compared to SM+ control. We also compared the patient's serum and synovial fluid against similar samples from an untreated patient. The results were similar, displaying diminished plaque formation in the patient's samples as compared to those from unexposed serum and synovial fluid (**Supplementary Fig. 2a, b**). Comparable results were observed with Φ83 changes in activity from neautralizing antibodies observed in patient serum (P < 0.001; **Fig. 3g**) or synovial fluid (P < 0.0005, **Fig. 3h**), but not in unexposed serum and synovial fluid (**Supplementary Fig. 2c, d**). These findings reaffirm the presence of neutralizing antibodies within both serum and synovial fluid samples, displaying a consistent and marked reduction in viral activity over time when challenged with Φ83 and PASA16. This response pattern suggests bacteriophage therapy maintained an active suppression of the infection in the presence of an immunological response from the patient's neutralizing serum and synovial antibodies.

Bacteriophages reduce patient *P. aeruginosa* biofilm

Control of the biofilm associated implant infection suggested the bacteriophages used had biofilm activity. We hypothesized that bacteriophage therapy could reduce the formation of *P. aeruginosa* biofilm. We saw a notable antibiofilm activity of bacteriophages Φ83 and PASA16 against *P. aeruginosa* clinical isolates collected at different time points. Utilizing both inhibition and eradication assays, concurrent incubation with these bacteriophages led to a significant reduction in biofilm formation compared to non-treated controls. In the inhibition assay, a substantial decrease in biofilm formation was observed. Pre-phage isolates (S1, S2), mid-term isolates (S8, S13) and late treated isolates (S22, S23) treated with PASA16 and Φ83 resulted in a significant decrease in biofilm formation (P=0.0001; **Fig. 4a,** Additional isolates in **Supplementary Fig. 3a).** Our results were similar when these experiments were performed with antibiotic treatment alone **(Supplementary Fig. 9)**.

Biofilms have a high tolerance to antibiotics[21]. Based on these results, we hypothesized that bacteriophages may have antibiofilm activity. Biofilm was cultured using the clinical isolates, treated with bacteriophage, and remaining biofilm was quantified. Pre (S1, S2), mid-term (S8,13), and late treated (S22, S23) isolates had a significant decrease in established biofilm following treatment as compared to untreated controls (P<0.01; **Fig. 4b,** additional isolates in **Supplementary Fig. 3b)**. These results demonstrate a robust capability of both bacteriophages in inhibiting and decreasing biofilm of *P. aeruginosa* isolates.

Bacteriophage therapy reduces *P. aeruginosa* disease severity and symptoms

To evaluate the efficacy of bacteriophage therapy, we focused on comparing isolates in the presence of bacteriophages, as the response was lost once bacteriophages were removed. We performed a *Caenorhabditis elegans* (*C. elegans)* infection model comparing clinical isolates in the presence and absence of bacteriophage. *C. elegans* infected with clinical isolates and treated with PASA16 and Φ83 showed a significant increase in survival compared to those infected with isolates not treated with bacteriophage (**Fig. 5a, 5b)**. This suggested that bacteriophages were either altering disease severity and symptoms or directly killing the bacteria.

To distinguish between these changes in virulence or cell death, these experiments were repeated with isolates resistant to bacteriophages. A S1 isolate resistant to PASA16 or Φ83, termed S1PASA16^r and S1Φ83^r, was generated by exposing the bacteria suspension of patient isolate S1 to top agar containing either PASA16 or Φ83. The calculated rates of *in vitro* phage resistance were (1.19 ± 0.91) x 10^5 CFU/mL for ΦPASA16 and (1.73 ± 1.47) x 10^5 CFU/mL for Φ83. *C. elegans* infected with the *in vitro* generated S1-resistant isolates (S1PASA16^r and S1Φ83^r) exhibited significantly increased survival compared to those infected with the parental patient isolate S1 that had not been exposed to bacteriophage (**Fig 5c**; **Supplementary Fig. 4a-f)**. This suggested that the presence of bacteriophages altered bacteria virulence as defined by disease severity and symptoms even when the bacteriophages were unable to kill the bacteria. Whole genome sequencing (WGS) of S1Φ83^r revealed a few notable mutations in glycosyl transferase *rfaB* and lipopolysaccharide biosynthesis protein GumC showed evidence of rearrangement. In S1PASA16^r, the *copB* gene, which encodes a copper resistance protein, was disrupted. These changes may indicate a trade-off between phage resistance and reduced virulence factor expression.

Swarming and swimming motility assays were completed to observe phage associated phenotypic changes on viable bacteria and confirm these results. Different isolates were not directly compared as this phenotype was lost during propagation in the clinical microbiology prior to research analysis. Clinical isolates treated with PASA16 and Φ83 showed a significant reduction in swarming (**Supplementary Fig. 5a-f**), swimming (**Supplementary Fig. 6a-f**) and twitching motility (**Supplementary Fig. 7a-f**) as compared to non-treated isolates. Results were similar when these experiments were completed with the phage resistant strains (S1PASA16^r and S1Φ83^r), confirming that the phenotypic effects depended on the direct presence of phage and was not only a result of bacteria lysis (**Fig. 5d-g**).

Genomic analysis of patient *P. aeruginosa* isolates

Given the patient's continued positive cultures but overall control of the infection, genomic variation was assessed to quantify any genotype changes in the *pseudomonas* isolates. Selective pressures from different local environments of the infection can drive adaptive mutations[22]. Based on pre-and post-treatment with phages and antibiotic susceptibilities patterns (**Supplementary Table 6**), twenty-seven isolates (pre-phage: S1, S2, S3, S4, S5, S6 and S7; long-term treatment: S8, S9, S10, S11, S12, S13, S14, S15, S16, S17, S18, S19, S20, S21, S22, S23, S24, S25, S26 and S27) were selected for whole genome sequencing (WGS). Each longitudinal isolate was compared with the assembled and annotated genome of the first patient isolate to identify mutations arising during the infection and/or associated with phage therapy (**Table 1**). Nineteen of these isolates were genetically identical to the initial isolate, whereas eight isolates acquired 1-3 new mutations of unknown relevance to the infection or phage treatment. These findings demonstrate limited evolution of the *P. aeruginosa* population during this infection that was controlled by bacteriophage therapy (**Table 2**). These results also showed that mutations and pseudogenes occurred in pre-mid and post-phage treated samples, but these mutations were not caused by the post-phage treatment and are not consistently present in post-phage treated samples (as shown in **Supplementary Fig. 8a, b**). WGS data revealed that phages were present on circularized contigs, and the read coverage profile is inconsistent with that of the bacterial genome. Therefore, it indicates that phage genes were present on aspiration.

Discussion

Antibiotic resistance is a public health emergency. Multi-drug resistant medical device-related infections illustrate these complex challenges. Here we demonstrate that bacteriophage therapy alone could be used to chronically suppress a MDR *P. aeruginosa* infection that had failed multiple surgical and antibiotic courses. Clinical control of the infection was established, in part, by bacteriophage modulating the disease severity and symptoms.

These types of orthopaedic implant and medical device infections with MDR Gram-negative organisms represent infections that do not have a practical antibiotic solution. The infections involve implants that are unable to be safely removed. Adequate oral antimicrobial options are not available to suppress the residual antibiotic tolerant biofilm that remains on the retained device[20, 22]. Additionally, it is not practical or safe to continue intravenous antimicrobial therapy in the overwhelming majority of these patients, given risks of adverse events and eventual development of antimicrobial resistance. We demonstrate that bacteriophage therapy was able to replace antibiotics as a long-term solution for chronic suppression of a periprosthetic joint infection, with no significant adverse events or development of resistance over an extended time period. Other reports have noted possible synergy between antibiotics and bacteriophage therapy[23], again emphasizing the importance of conducting clinical studies despite the known hurdles.

When bacteriophage therapy was used as the sole antimicrobial agent, a phenotypic decrease in *P. aeruginosa* virulence resulted in clinical control of the infection without its eradication. This was defined as the complete or significant reduction of infection-related symptoms, relative normalization of laboratory markers, and the absence of a need for further antimicrobial or surgical interventions. This was observed by a statistically significant increase in *C. elegans* survival and reduced motility and swarming after treatment with bacteriophage. Our WGS of phage resistance isolates revealed mutations in genes related to lippolysaccharide (LPS) biosynthesis and cell envelope function, especially in *rfaB*, *GumC*, and *copB*. Previous research has demonstrated that changes in LPS-related genes are common mechanisms of phage resistance, as they affect phage receptor recognition[24,25]. Importantly, these modifications are often associated with fitness trade-offs, including reduced virulence. Our findings also showed that the observed mutations may confer phage resistance while simultaneously attenuating virulence, as indicated by increased survival in the C. *elegans* infection model.

In both the *in vitro* experiments and the clinical treatment, these observed changes in the bacteria phenotype occurred in the presence of bacteriophage. These changes were not from host adaptation. The pre-phage isolates were an already host-adapted clone that colonized the patient for several years. As well, there was minimal genotype variation between clinical *Pseudomonas* isolates before and after phage treatment. Phage therapy inhibited biofilm formation, likely due to the lysozyme-like endolysin, which degrades the peptidoglycan layer of bacterial cells, leading to cell lysis. The lysis of cells embedded within the biofilm may weaken its structural integrity and reduce viability. However, the exact molecular mechanism of this anti-biofilm activity remains unclear. Host antibodies were observed directed to the bacteriophage, but continued clinical utility continued to be observed. This has been reported in other case reports[26,27].

The presence of neutralizing antibodies in synovial fluid and serum demonstrates a host immune response. This finding aligns with previous studies that have shown mucosal and tissue-localized antibody responses can directly impact phage viability and therapeutic effect, particularly when phages are repeatedly administered or circulate for extended periods[28]. Comparative analyses of serum and synovial fluid from an untreated patient confirmed that the observed neutralizing effects were treatment-specific, as unexposed control samples retained full lytic activity. The therapeutic effects of this observation remain unknown until well-designed clinical studies can be completed.

We demonstrate that bacteriophage therapy was able to replace antibiotics as a long-term solution for chronic suppression of a periprosthetic joint infection, with no significant adverse events or development of resistance over an extended time period. Other reports have noted possible synergy between antibiotics and bacteriophage therapy, again emphasizing the importance of conducting clinical studies despite the known hurdles.

To date, bacteriophage therapy has limited clinical data proving effectiveness for different infectious disease conditions. However, case series suggest there is a signal of efficacy[10], but no randomized prospective studies have been successfully completed. Multiple groups have attempted to conduct phase 1 and 2 clinical studies and have failed. These include a Phase 2 study for chronic salvage PJI (PhageBank, NCT05269134) and a phase 2 study for acute PJI cases (GLORIA, NCT06605651). These studies had challenges with enrollment, logistical issues, and concerns with economic reimbursement. More recently, other groups have reported successful completion of bacteriophage phase 2 clinical studies in diabetic foot osteomyelitis (BX211, NCT05177107). These studies enrolled patients in the United States with one in the European Union. This highlights the difficulties and potential of conducting randomized controlled studies with bacteriophage therapy in complex clinical diseases that have limited antimicrobial and surgical options across different countries and regulatory agencies. Based on these limitations, bacteriophage therapy has not obtained regulatory approval and can only be used in expanded access cases. This work serves as a proof of concept for the utility of bacteriophage therapy in these difficult medical device related infections that do not have a surgical and antibiotic solutions. Given these challenges, it adds additional evidence to the argument that these complex cases should be managed at designated centers of excellence for periprosthetic joint infection that possess the microbiology, regulatory, and clinical expertise[24, 25].

These findings suggest that the use of bacteriophage therapy resulted in the clinical control of the infection without eradication secondary to phage biofilm activity and a phenotypic alteration in bacterial virulence. This demonstrates an important clinical concept for the ability of bacteriophage to be used in chronic suppression of implant-associated infections. When an antibiotic option does not exist with multidrug-resistant bacteria, our data provides preliminary evidence that bacteriophage may be an alternative therapy. This supports conducting prospective clinical studies.

Methods

Ethical statements

The patient has consented to our use of samples and data in research and to the publication of clinical data, the images presented here. They also gave written informed consent to receive phage therapy in accordance with CARE guidelines. To establish bacteriophage therapy, patient cultures were sent for bacteriophage matching (Adaptive Phage Therapeutics, Bethesda, MD) and a phage match was obtained. Two bacteriophages were identified with a high activity (PASA16 and Φ83). We obtained compassionate-use IRB (EA21110067) from the University of Pittsburgh and FDA approval under expanded access (IND 28197). Bacteriophages were screened using Adaptive Phage Therapeutics bioinformatics pipeline to ensure the absence of lysogeny-associated genes, virulence factors, and antimicrobial resistance genes for its phage repository. The screening results were submitted by APT to the U.S. Food and Drug Administration (FDA) as part of its drug master file in support of applications for expanded access and clinical trials.

MSIS criteria

The patient was included in the study based on a confirmed diagnosis of PJI using the MSIS 2013 criteria[29]. These criteria were applied to ensure accurate and standardized identification of infection. Specifically, patient met either one of the major MSIS criteria—such as a sinus tract communicating with the prosthesis or isolation of the same microorganism from at least two separate periprosthetic specimens or at least three of the minor criteria, which included elevated inflammatory markers (CRP, 10 mg/L and

ESR, 30mm/hr), increased synovial WBC count (3,000 cells/mL) or polymorphonuclear % (PMN, 80%), positive histological findings, or a single positive culture result. The patient also met the comparable definition of PJI of International Consensus Meeting (ICM) 2018 based on these same criteria[18]

Bacterial strains, bacteriophage and growth conditions

Synovial fluid and blood samples were collected from the patient during clinical visits. Serum was extracted from blood samples and stored as frozen aliquots at – 80 °C. *P. aeruginosa* isolates, collected from the patient's cultures, were provided by the UPMC Presbyterian Microbiology Lab. Bacteriophages, Φ83 and PASA16 (Accession number: MT933737.1), their host cell, C10, and the original patient cultures, S1 and S2, were provided by Adaptive Phage Therapeutics (APT). Φ83 was a proprietary bacteriophage provided under a material transfer agreement from APT and the United States Department of Defense. Its genome is not publicly available. All bacterial strains were grown in Mueller Hinton Broth (MHB; Bectin Dickinson and Company) medium with shaking at 37 °C and stored as frozen stocks in 30% glycerol at – 80 °C. All phages were tested using a standard plaque assay, expanded, purified and stored as frozen stocks in 30% glycerol at – 80 °C.

Growth assay

The clinical isolates were inoculated and cultured on blood agar plates from a -80°C stock. The cultures were then transferred into 5 mL of MHB medium and incubated overnight at 37°C. After incubation, the cultures were standardized to 5×10^5 CFU/mL using a 0.5 McFarland turbidity standard (Hardy Diagnostics). The standardized cultures were plated in triplicate in 96-well plates (Costar) and the optical density was measured in every 30 minutes for 24 hours at a wavelength of 600 nm using a microplate reader (BioTek).

Growth inhibition assay

S1, S2, S8, S13, S22, and S23 were inoculated into 5 ml MHB medium and grown overnight at 37ºC. Cultures were normalized to $5\text{x}10^5$ CFU/mL using a 0.5× McFarland turbidity standard (Hardy Diagnostics). A multiplicity of infection (MOI) of approximately 10 was used for bacteriophages PASA16 and Φ83 during the plaque assay. Cultures were plated in triplicates in 96 well plates (Costar) and bacteriophages were added and incubated at 37 °C for 24 h in a shaking incubator. After 24 h the optical density was observed in the microplate reader at 600nm (BioTek).

Plaque Assay

To perform the experiment, 100 µL of each clinical isolate's overnight culture was mixed with 10mL of pre-warmed lysogeny broth (LB) containing 0.75% (w/v) of agar (molten agar) and 5mM of $CaCl_2$. This mixture was then poured onto warmed bottom agar plates (1.5% agar) and allowed to solidify. Once the agar had solidified, 10µL drops of PASA16 and Φ83 phage stock, which had been serially diluted ten times, were placed on top of the agar. The drops were left to dry at room temperature and the Petri dishes were then incubated at 37°C overnight. The next day, the plates were checked for lysis zones and the PFU were calculated.

Minimum inhibitory concentration (MIC)

The minimum inhibitory concentrations (MICs) of cefepime, ciprofloxacin, meropenem, and Zosyn for all patient cultures were determined by the UPMC Presbyterian Microbiology Lab. The MicroScan lyophilized broth microdilution system (Beckman Coulter, Brea, CA, USA) was used for MIC testing as per the manufacturer's guidelines.

Phage-targeting antibody neutralization assay

To assess phage stability in serum, high-titer lysates of Φ 83 and PASA 16 were prepared. Patient serum samples from pre-phage and post-phage time points were thawed on ice. Briefly, 180 µL of high-titer phage stock (approximately 1×10^9 PFU/mL) was mixed with 20 µL of serum (9:1) and incubated at room temperature overnight. After the incubation, 20 µL from each mixture was serially diluted in SM+ buffer. Then, 5 µL aliquots of each dilution were spotted in triplicate onto a top agar of C10. SM+ buffer was used as a control. The agar plate was incubated at 37 °C for 24 h. Plaques were counted, and the presence of neutralizing activity was determined in PFU/mL. Additionally, to confirm the presence of neutralizing antibodies this assay was repeated to compare the patient's serum and synovial fluid samples to control human serum and synovial fluid samples.

Biofilm inhibition assay

S1, S2, S8, S13, S22, and S23 were inoculated into 5 ml MHB medium and grown overnight at 37ºC. Cultures were normalized to $1x10^5$ CFU/mL using a 0.5× McFarland turbidity standard (Hardy Diagnostics) and bacteriophages, Φ83 and PASA16, were diluted to approximately $1x10^8$ PFU/mL using a standard plaque assay. For a 200 µL reaction, we used 182 µL of bacterial culture was added to the 96-well plate, and 18 µL of phage was then added to achieve a multiplicity of infection (MOI) of 1:100. The experiment was conducted in triplicate with untreated bacteria as the control. The plate was incubated at

37 °C for 48 h. After incubation, non-adherent cells were removed by turning the plate over and shaking out the liquid. The plate was gently submerged in a small tub of water and then shaken out. Using a micropipette, 200 uL of water were gently added to the wells and then shaken out; this process was repeated 3-4x. Next, 125 uL of crystal violet (CV) was added to the wells and incubated at room temperature for 10-15 minutes. The wells were gently rinsed 3-4x and left to dry upside down in laminar flow hood for 10-15 minutes. To solubilize the CV, 125 uL of 30% acetic acid was added to the wells and incubated at room temperature for 10-15 minutes. To assess biofilm inhibition, absorbance was quantified in a plate reader at 550 nm using 30% acetic acid as the blank. For qualitative purposes, plates were photographed.

Biofilm eradication assay

S1, S2, S8, S13, S22, and S23 were inoculated into 5 mL MHB medium and grown overnight at 37ºC. Cultures were normalized to $1x10^5$ CFU/ml using a 0.5× McFarland turbidity standard (Hardy Diagnostics) and bacteriophages, Φ83 and PASA16, were tittered to approximately $1x10^8$ CFU/ml using a standard plaque assay. For a 200 μL reaction, we used 182 μL of bacterial culture was added to the 96-well plate, and 18 μL of phage was then added to achieve a multiplicity of infection (MOI) of 1:100. The experiment was conducted in triplicate with untreated bacteria as the control. Cultures were incubated at 37 °C for 48 h. After 48 h, Φ83 and PASA16 were added to the wells. The plate was incubated at 37 °C for 28 h. After incubation, non-adherent cells were removed by turning the plate over and shaking out the liquid. The plate was gently submerged in a small tub of water and then shaken out. Using a micropipette, 200 uL of water were gently added to the wells and then shaken out; this process was repeated 3-4x. Next, 125 uL of crystal violet (CV) was added to the wells and incubated at room temperature for 10-15 minutes. The wells were gently rinsed 3-4x and left to dry upside down in laminar flow hood for 10-15 minutes. To solubilize the CV, 125 uL of 30% acetic acid was added to the wells and incubated at room temperature for 10-15 minutes. To assess biofilm eradication, absorbance was quantified in a plate reader at 550 nm using 30% acetic acid as the blank. For qualitative purposes, plates were photographed.

Biofilm formation in patient-derived *Pseudomonas aeruginosa* isolates treated with antibiotics

Biofilm formation was evaluated in patient-derived isolates of *Pseudomonas aeruginosa* following antibiotic treatment. Untreated cultures served as baseline controls and were compared with cultures treated with cefepime, ciprofloxacin, meropenem, and tazobactam/piperacillin. Isolates were categorized based on the stage of treatment: pre-phage treatment (S1, S2), mid-treatment (S8, S13), and late-stage treatment (S22, S23). Cultures were normalized to $1x10^5$ CFU/mL using a 0.5× McFarland turbidity standard, and 200 μL was added to 96-well plate. Each isolate was exposed to antibiotics at concentrations corresponding to 5x, 20x, and 50x their respective MICs. The cultures were incubated under static conditions for 48 hours in 96-well plates. After incubation, wells were gently washed to remove planktonic cells, stained with crystal violet, and the biofilm biomass was quantified by measuring the OD at 600 nm

In vitro phage resistance assay

Both the phage resistance experiments were performed in MHB (Bectin Dickinson and Company). A phage suspension in top agar was prepared by adding 1 x 10^8 PFU of either ΦPASA16 or Φ83 to molten top agar (0.7% agar) which was then plated onto agar plates (1.5% agar) and allowed to solidify. The clinical isolate S1 was grown to the exponential phase and then serially diluted 1:10 in sterile PBS across 10 dilution steps. Five microliters of each dilution were then spotted onto the solidified phage suspension, allowed to dry and then incubated overnight at 37°C. The next day, CFUs were counted at dilution steps. For each phage suspension plate, the number of countable CFUs was divided by the total CFUs as estimated by the negative control calculations. This calculation provided a "resistance rate" indicating the estimated number of CFUs that contained at least 1 phage-resistant mutant.

To reconfirm phage resistance, a cross-streak assay was conducted using colonies isolated from the phage suspension plates. First, over 10 potentially resistant colonies were selected and grown overnight in fresh media. The next day, $1x10^8$ PFU of either PASA16 or Φ83 was streaked across the center of the agar plate and allowed to dry. Then, 10 microliter samples from each colony's overnight culture or S1 were spotted onto the agar above the phage line. The plates were incubated overnight at 37°C. The following day, strains that were phage-susceptible and phage-resistant were identified (Supplementary Fig. 10).

***C. elegans* maintenance**

The wild type *C. elegans* (N2 strain) was purchased from Carolina Biological Supply in Burlington, NC, USA for use in this study. The nematodes were grown and propagated on nematode growth media, which consisted of 15g agar, 2.4g $NaCl_2$, 2g tryptone, 2.72 g KH_2PO_4, 0.8 mL of 1 M $CaCl_2$, 5mg/mL of cholesterol, and 1 M $MgSO_4$ per L. The nematodes were fed with *E. coli* OP50 and kept at 25^OC[30]. To synchronize the worms, 5M NaOH and 5% bleach were used, and eggs were collected. The eggs were washed with M9 medium, which contains 6g Na_2HPO_4, 3g KH_2PO_4, 5g NaCl, and 0.25 g $MgSO_4 \cdot 7H_2O$. The eggs were then incubated with S-complete medium (S-basal medium: 5.9 g NaCl, 50 mL of 1M potassium phosphate, pH 6.0 in 1L; S-complete medium: 977 mL S-basal, 10 mL 1M potassium citrate pH 6.0, 10 mL Trace metals solution, 3 ml 1M $CaCl_2$, 3 mL 1M $MgSO_4$) to allow hatching. For the experiment, age-synchronized L4 worms were incubated in 96-well plates containing 100µL of S-complete medium in each well.

***C. elegans in vivo* infection model**

For the infection assay, 96-well plate was filled with S-complete medium added with overnight culture of *E. coli* OP50. Then 15 mature L4 stage *C. elegans* were transferred in to medium. The total volume in the 96 well plate was maintained at 100 µL. Next, clinical isolates (*P. aeruginosa*) were grown overnight and washed with PBS and maintain the $1x10^6$ CFU/mL for infection. The efficacy of phage therapy was studied using a concentration of bacteria and phage with 1:100 multiplicity of infection (MOI). The plates were incubated at 20°C, and survival was monitored every 24 h for 7 days. The results were observed and evaluated based on live and dead *C. elegans*. All the experiments were repeated a minimum of three times for statistical significance.

Table 3. Experimental setup for *C. elegans* in vivo infection model

Groups	***P. aeruginosa* clinical isolate ($1x10^6$ CFU/mL)**	**PASA16 (PFU/mL)**	**Φ83 (PFU/mL)**
Group 1 (No infection control)	-	-	-
Group 2 (infection control)	S1	-	-
Group 3 (treatment)	S1	$1x10^8$	-
Group 4 (treatment)	S1	-	$1x10^8$
Group 5 (infection control)	S2	-	-
Group 6 (treatment)	S2	$1x10^8$	-
Group 7 (treatment)	S2	-	$1x10^8$
Group 8 (infection control)	S8	-	-
Group 9 (treatment)	S8	$1x10^8$	-
Group 10 (treatment)	S8	-	$1x10^8$
Group 11 (infection control)	S13	-	-
Group 12 (treatment)	S13	$1x10^8$	-
Group 13 (treatment)	S13	-	$1x10^8$
Group 14 (infection control)	S22	-	-
Group 15 (treatment)	S22	$1x10^8$	-

Group 16 (treatment)	S22	-	1x10^8
Group 17 (infection control)	S23	-	-
Group 18 (treatment)	S23	1x10^8	-
Group 19 (treatment)	S23	-	1x10^8
Group 21 (infection control) (S1PASA16^r)	S1PASA16^r	-	-
Group 22_ (infection control) (S1Φ83^r)	S1Φ83^r	-	-

Motility assays

Swarming Motility Assay

Swarming motility was tested as reported previously with a few modifications[31]. To create the swimming plates, 6.5 g of agar was added to 800 mL of water, and the mixture was then autoclaved. 200 mL of a 5× M8 solution was then added to the agar mixture. The 5× M8 solution was created in advance by combining 30 g Na_2HPO_4, 15 g KH_2PO_4, and 2.5 g of NaCl with enough water to obtain a final volume of 1 L. After the addition of the 5× M8 solution, 10 mL of 20% glucose, 25 mL of 20% casamino acids, and 1 mL of 1 M $MgSO_4$ were combined with the agar solution. The solution was mixed and cooled prior to being poured in petri dishes (～25 mL/plate), and the plates were allowed to solidify overnight at room temperature.

For the experimental test plates, 10 μL of an overnight bacterial culture from a clinical isolate was combined with 10 μL of the phage being tested (PASA16 or ϕ83). This step was repeated for each of the 6 clinical samples being tested. The plates were then inoculated with 2.5 μL of bacteria, or 2.5 μL of the bacteria and phage mixture, and incubated upright for 24 hours. Following incubation, the zone diameter was recorded for each plate.

Swimming Motility Assay

Swimming motility was analyzed as previously described with minor modifications[31]. Plates were created as previously described for the Swarming Motility Assay, but 3 g of agar was used. For the experimental test plates, 10 μL of an overnight bacterial culture from a clinical isolate was combined with 10 μL of the phage being tested (PASA16 or ϕ83). A toothpick was dipped into this mixture, and stabbed partway into the agar plates. The plates were then incubated upright for 24 hours, and the zone diameter was measured.

Twitching Motility Assay

Twitching motility was accessed as previously described with minor modifications[32]. LB agar plates were stab inoculated with a mixture of 10 μL of an overnight bacterial culture from clinical samples and 10 μL of the phage being tested (PASA16 or ϕ83). The plates were then incubated for 48 hours and the agar was removed. The remaining attached cells were stained with a 1% Crystal violet solution, and the size of twitching was then measured.

Phage-antibiotic growth inhibition assay

Pre-phage clinical isolate S1 was grown to the stationary phase in TSB and then diluted 1:100 into fresh media, allowing it to enter log-phase. The culture was then normalizing with additional fresh media to an OD of 0.1, which corresponds to 1×10^8 CFU/mL. Aliquots of the culture were treated with either 0.5xMIC of Pip/Tazo (Sigma-Aldrich, MO, USA), PASA16, Φ83, or a 50:50 cocktail at a MOI of 0.05; as well as combinations of these treatments. The plates were then incubated at 37°C with intermittent shaking in a multiplate plate reader, and the optical density was measured every 30 minutes. The assay was performed in triplicate.

Statistical analysis

All graphical and statistical analysis was performed using Prism 9.5 (GraphPad, La Jolla, CA). Statistical analysis was performed using one-way ANOVA with multiple comparisons test for the neutralizing antibodies data, and an unpaired t-test for the biofilm data. Significance was determined at $p < 0.05$. Images were treated with ImageJ (v1.53).

Whole-genome sequencing and Raw read processing

DNA from twenty-seven *P. aeruginosa* isolates was extracted using Qiagen kit according to the manufacturer's protocol. Briefly, genomic DNA was isolated from frozen cell pellets using the DNeasy Blood and Tissue Kit (Qiagen, Hilden, Germany) specifically tailored for the QIAcube. The elution process was conducted over a 10-minute period into nuclease-free water to ensure the purity of the DNA. The concentration of the extracted DNA was determined using the Qubit® 3.0 Fluorometer. Later, sample was sent to SeqCoast Genomics for further analysis. Trimmomatic (version 0.39) was used with the following parameters to quality-trim reads based on phred33 quality scores and remove adaptor sequences[33]. Genomes were assembled using the Unicycler software (version 0.4.4)[34]. Annotation was performed using BAKTA (version 1.6.1)[35]. Comparative genomics was performed using DIAMOND (version 2.0.15.153, Buchfink *et al*., 2015) and the Pseudofinder software[36]. Variant-calling was performed using the Breseq software (version 0.36)[37]. Breseq was run with the consensus-minimum-variant-coverage-each-strand flag set to 3, and all other default parameters. Sample 1 was used as the reference genome, and false-positive mutations were filtered out based on their presence in sample 1. Principal component analysis was performed using the R package *factoextra.*

Data Availability

Detailed clinical protocols, results and additional data are available in the paper and in the Supplementary file. The details about the patient and bacteriophage therapy are also included in the Supplementary information. The genome sequence of bacteriophage is accessible in the GenBank database under the accession MT933737.1. The *P. aeruginosa* genome sequence reads are deposited under NCBI BioProject PRJNA1406270. All other data supporting the findings of this study are available within the paper. Source data is provided with this paper. Requests for additional data, resources, and reagents should be directed to the lead contact, Kenneth Urish (ken.urish@pitt.edu). Reagents, if possible, will be made available to qualified researchers under a completed Materials Transfer Agreement and USDA and CDC permits, if required.

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Author contributions

All authors contributed to the data analysis and outlined the study; R.A. and K.L.U designed the experiments. R.A., D. M., N.M.S., A.J.F. and J.J.R. conducted most of the studies; V.S.C. conducted the bioinformatic analysis and analysed the data; R.A., K.L.U wrote the manuscript with contributions from other authors.

Competing interests

The authors declare no competing interests.

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Tables

Table 1. Chromosomal mutations identified in *Pseudom* *aeruginosa* isolates from patient before phage treatment

Gene/ Sample:	S1	S2	S3	S4	S5	S6	S7
ubiG	-	-	-	-	-	-	-
fabG	-	-	-	-	-	-	-
yqjQ	-	-	-	-	-	-	-
DUF1456	-	-	-	-	-	-	-
muiA	-	-	-	-	-	-	-
aroG1	-	-	C>T	-	-	-	-
wzy	-	-	-	-	-	-	G>T
dnaX	-	-	-	-	-	-	-

Table 2. Chromosomal mutations identified in *Pseudomonas aeruginosa* isolates from patient undergoing long-term phage treatment.

***Gene/* Sample:**	***dnaX***	***wzy***	***aroG1***	***muiA***	***DUF***	***yqjQ***	***fabG***	***ubiG***
S8	C>A	-	-	-	-	-	-	-
S9	-	-	-	-	-	-	-	-
S10	-	-	-	-	-	-	-	-
S11	-	-	-	-	-	-	-	-
S12	-	-	-	-	-	-	-	-
S13	-	-	-	-	-	-	-	-
S14	-	-	-	-	-	-	-	-
S15	-	-	-	-	-	-	-	-
S16	-	-	-	-	-	-	-	-
S17	-	-	-	-	-	-	-	-
S18	-	-	-	-	-	-	-	-
S19	-	-	-	-	-	-	-13	G>A
S20	-	-	-	-	-	-	-	-
S21	-	-	-	-	-	-	-	-
S22	-	-	-	C>T	A>T	C>G	-	-
S23	-	-	-	-	-	-	-	-
S24	-	-	-	-	-	G>C	-	-
S25	-	-	-	-	A>T	G>C	-	-
S26	-	-	-	-	-	-	-	-
S27	-	-	-	-	-	-	-	-

Figure legends

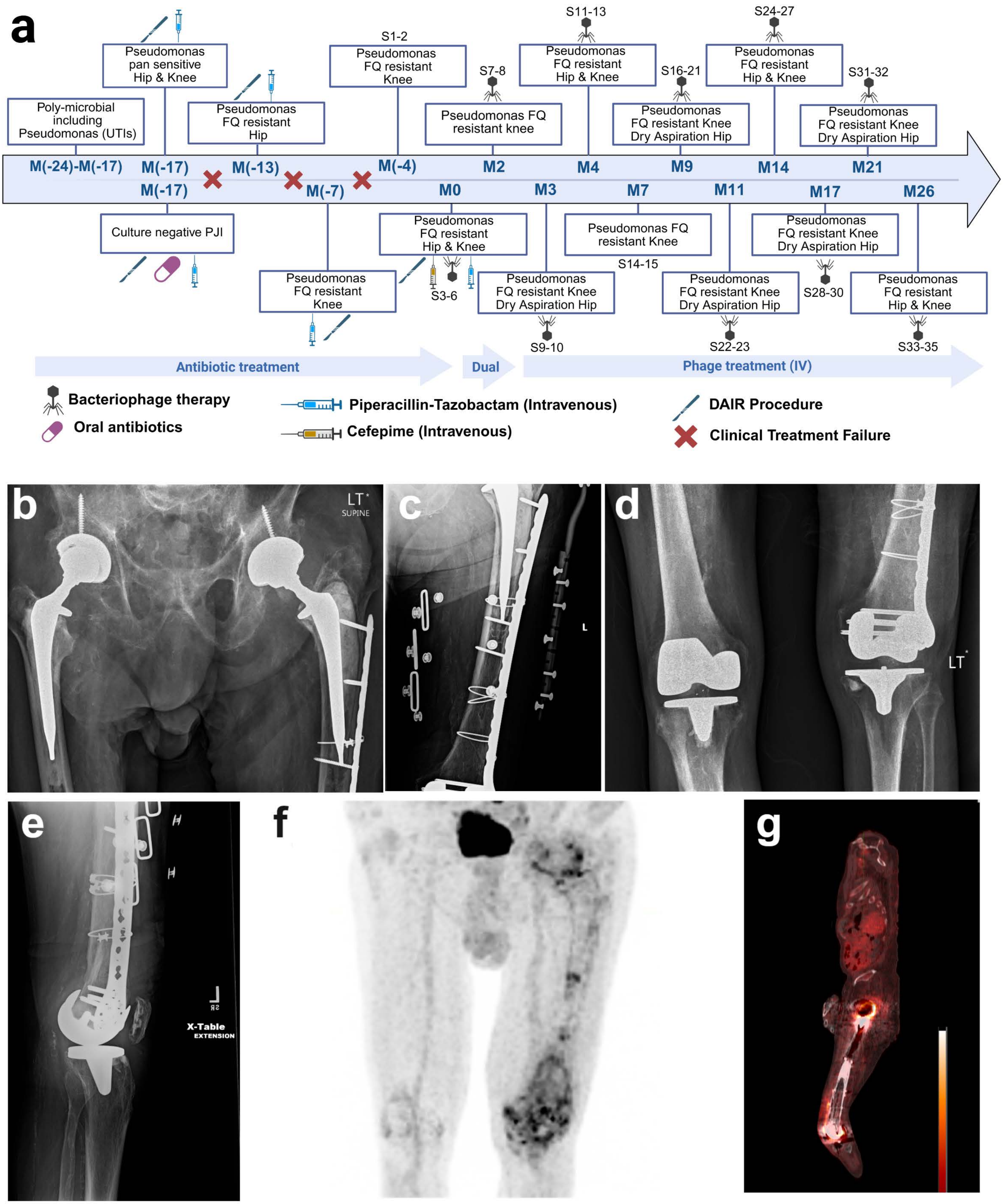
Figure. 1. *Clinical Timeline and Imaging.* ***a,*** *Clinical timeline of PJI course by months. M0 designates time of first bacteriophage therapy. Bacteriophage therapy (black), oral antibiotic treatment (purple), intravenous* piperacillin-tazobactam *(blue), intravenous cefepime (yellow), fluoroquinolone* (*FQ). Created in BioRender. Arya, R. (2026) https://BioRender.com/d9m0820. Radiographs demonstrate complexity of implants and hardware.* ***b,*** *Anterior-posterior supine pelvis radiograph;* ***c****, Left lateral femur radiograph;* ***d****, Anterior-posterior standing knee radiograph;* ***e****, Lateral left knee extension radiograph; PET imaging demonstrated PJI of entire femur:* ***f****, Anterior-Posterior PET imaging of the pelvis and upper leg;* ***g****, Lateral PET-CT of left upper.*

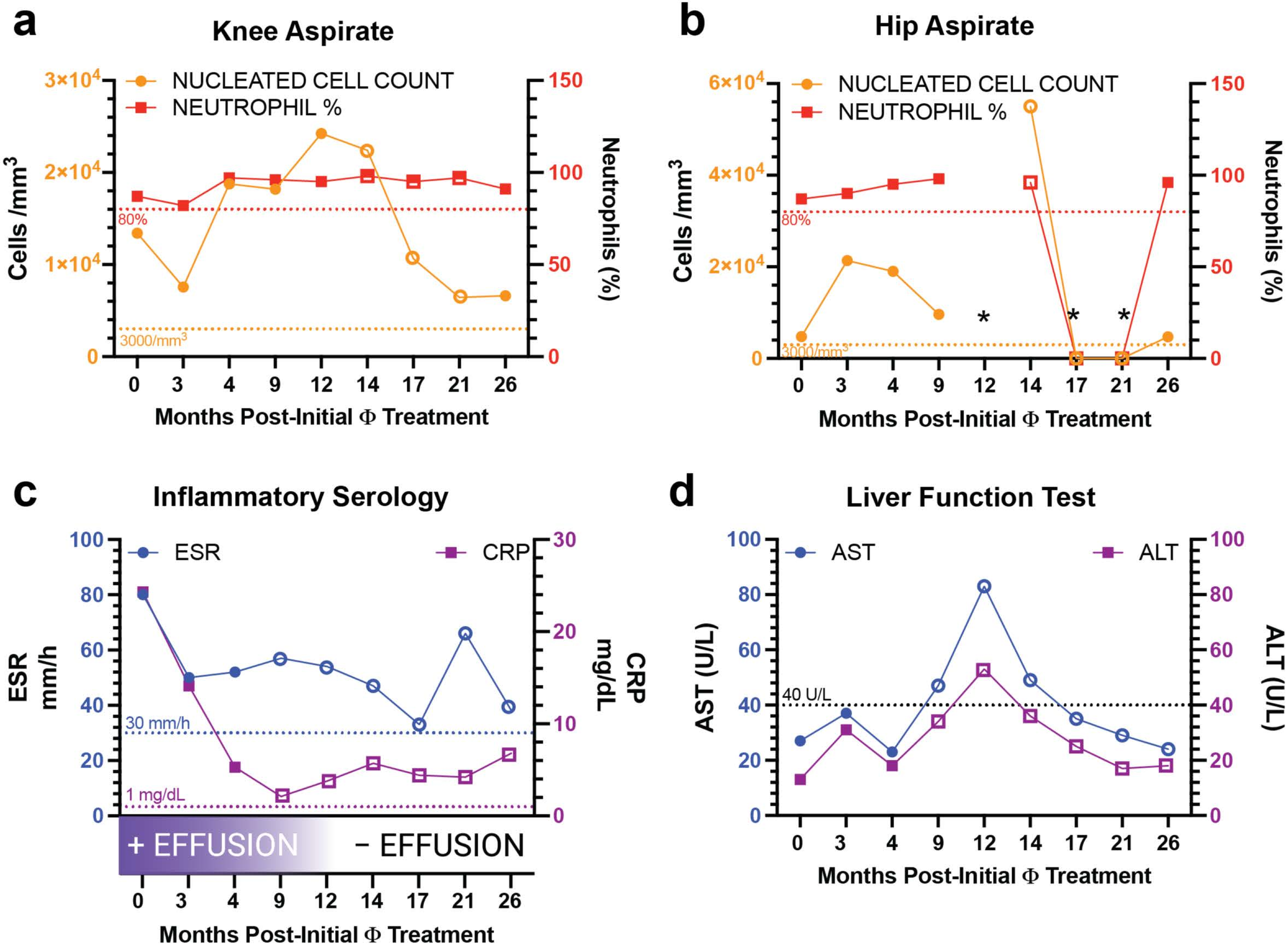
Figure 2. Clinical laboratory values measured at different time points (0, 3, 4, 9, 12, 14, 17, 21 and 26 months) following the initiation of phage therapy. ***a****, knee aspirate with nucleated cell counts and differential; half-shaded orange circles and red squares indicate moderate or minimal effusion beginning at 14-months post initial phage therapy.* ***b****, hip aspirate data; the asterisks, open orange circles and red squares show the absence of effusion.* ***c****, inflammatory serology; erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) levels, with open blue circles and purple squares representing no reported pain by the patient, starting after the 12-months. Effusion resolved at 6-9 months post-initial treatment as represented by the shading.* ***d****, liver function displaying AST and ALT levels, with thresholds marked at ≤ 40 U/L for normal values. Dotted horizontal lines in all graphs represent the elevated threshold for Musculoskeletal Infection Society (MSIS) and International Consensus Meeting (ICM 2018) criteria for prosthetic joint infection (PJI) including nucleated cell count <3000 cells/mm³, neutrophils <80%, ESR 0-23 mm/h, and CRP <0.8 mg/dL.*

Figure. 3. *Phage lytic activity on pre-, mid- and late-term treatment samples**.** **a**, PASA16 visualized by TEM**.** **b**, Φ83 Visualized by TEM**.** Planktonic growth inhibition by PASA16 and Φ83 (MOI 1:10) on clinical isolates. **c**, Inhibition of planktonic growth by PASA16 on Pre- mid and late-term treated isolates. **d**, Inhibition of planktonic growth by Φ83 on Pre- mid and late-term treated isolates. The presence of neutralizing antibodies against bacteriophages PASA 16 and Φ83 in patient serum and synovial fluid samples. **e,** PASA16 activity in SM+ buffer and serum samples from different time points (6, 9, 12 and 24 months). **f**, PASA16 activity in synovial fluid sample. **g**, Φ83 activity in SM+ buffer and serum samples from different timepoints (6, 9, 12 and 24 months). **h**, Φ83 activity in synovial fluid sample. Values are presented as mean ± SEM of three independent experiments performed in technical replicates.*

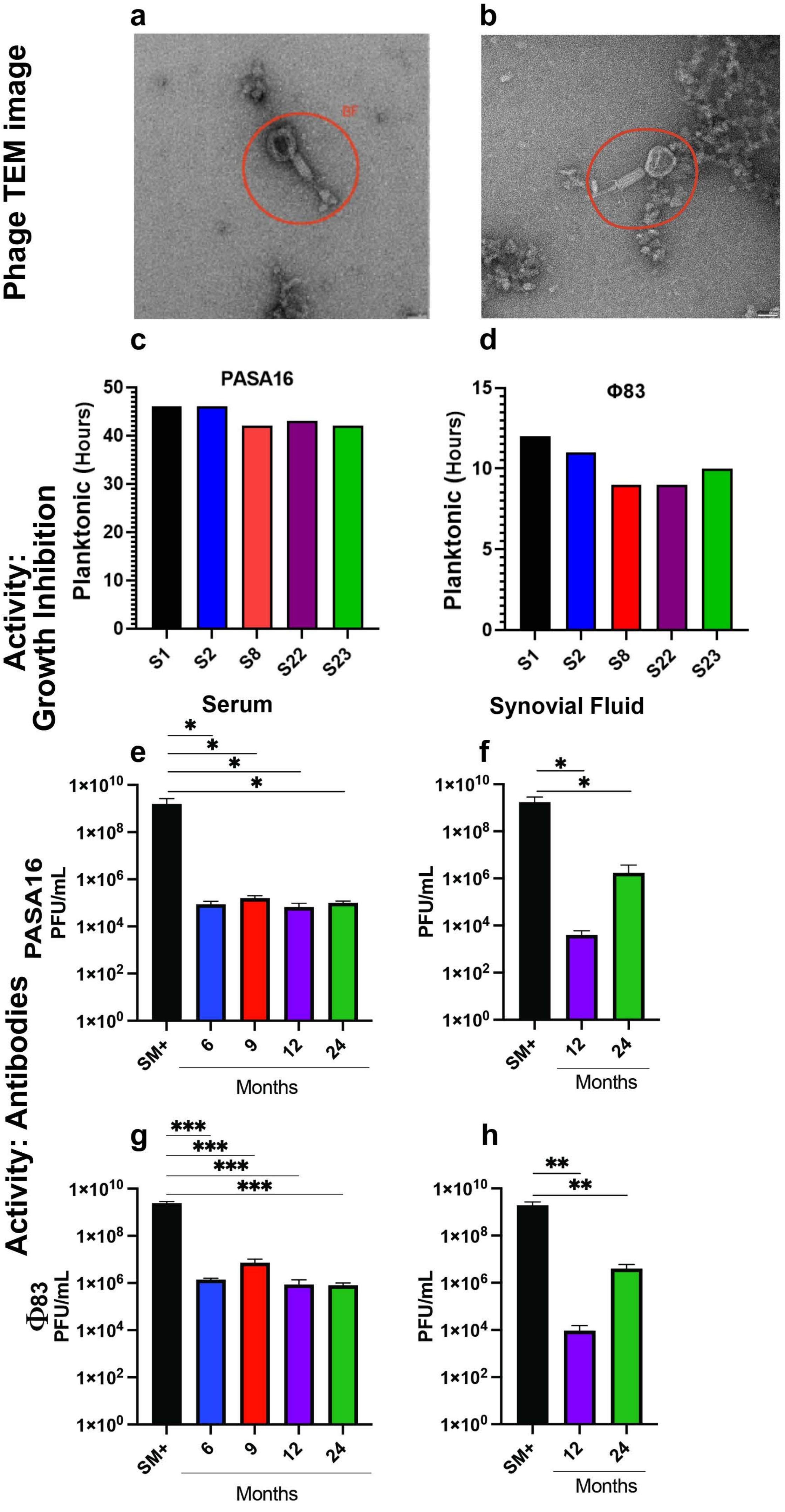
Figure. 4. *Biofilm formation and eradication in patient-derived Pseudomonas aeruginosa isolates. **a**, PASA16 and Φ83 inhibiting P. aeruginosa biofilms (bacteria and phage ratio of 1:100, i.e., 10^6 CFU/mL and 10^8 PFU/mL), in the pre-phage sample (S1, S2), mid-term (S8, S13) and late-term treatment samples (S22 and S23). **b**, PASA16 and Φ83 phages in eradicating P. aeruginosa biofilms (bacteria and phage ratio of 1:100, i.e., 10^6 CFU/mL and 10^8 PFU/mL) in the pre-phage sample (S1, S2), mid-term (S8, S13) and late-term treatment samples (S22 and S23). Photographic evidence of the Crystal Violet-stained biofilms provides qualitative validation. Experiments were performed in replicates.*

Figure. 5. *Phage therapy in C. elegans infected with patient isolates and compared with in vitro-evolved phage resistant strains. **a**, Schematic representation of experimental design. Created in BioRender. Arya, R. (2026) https://BioRender.com/q3oo343.; **b,** Representative survival of C. elegans infected by pre-, mid- and long-term samples and treated with PASA16 and Φ83 at an MOI of 100; **c,** Survival of C. elegans infected with phage resistant strains evolved from S1 to resist PASA16 and Φ83* (*S1PASA16$^{\mathbf{r}}$ and S1Φ83$^{\mathbf{r}}$*). e*ffect of bacteriophage on swarming motility in P. aeruginosa clinical isolates. **d**, Pre-, mid- and long-term isolates were treated with PASA16, Φ83 and swarming was analysed after 16 h at 37°C. Zone diameter were measured in millimetres. **e,** Swarming was assessed in the resistant strains (S1PASA16$^{\mathbf{r}}$ and S1Φ83$^{\mathbf{r}}$). Effect of bacteriophage on swimming motility in P. aeruginosa clinical isolates**.** **f,** Pre- mid- and long-term isolates were treated with PASA16, or Φ83 and swimming were analysed after 16 h at 37°C and zone diameter was measured in millimetres. **g**, Swimming was analyzed* *in resistant strains (S1PASA16^{r} and S1Φ83^{r}). All experiments were performed in triplicate and asterisks indicate significant differences between means at $p < 0.05$ by analysis of variance followed by unpaired t-test.*


a
Poly-microbial including Pseudomonas (UTIs)
Pseudomonas pan sensitive Hip & Knee
Culture negative PJI
Pseudomonas FQ resistant Hip
Pseudomonas FQ resistant Knee
S1-2
Pseudomonas FQ resistant Knee
Pseudomonas FQ resistant Hip & Knee
S3-6
S7-8
Pseudomonas FQ resistant knee
Pseudomonas FQ resistant Knee Dry Aspiration Hip
S9-10
S11-13
Pseudomonas FQ resistant Hip & Knee
Pseudomonas FQ resistant Knee
S14-15
S16-21
Pseudomonas FQ resistant Knee Dry Aspiration Hip
Pseudomonas FQ resistant Knee Dry Aspiration Hip
S22-23
S24-27
Pseudomonas FQ resistant Hip & Knee
Pseudomonas FQ resistant Knee Dry Aspiration Hip
S28-30
S31-32
Pseudomonas FQ resistant Knee Dry Aspiration Hip
Pseudomonas FQ resistant Hip & Knee
S33-35
M(-24)-M(-17)
M(-17)
M(-17)
M(-13)
M(-7)
M(-4)
M0
M2
M3
M4
M7
M9
M11
M14
M17
M21
M26
Antibiotic treatment
Dual
Phage treatment (IV)
Bacteriophage therapy
Oral antibiotics
Piperacillin-Tazobactam (Intravenous)
Cefepime (Intravenous)
DAIR Procedure
Clinical Treatment Failure
b
LT
SUPINE
c
L
d
LT
e
X-Table
EXTENSION
f
g



a
Knee Aspirate
NUCLEATED CELL COUNT
NEUTROPHIL %
Cells /mm3
Neutrophils (%)
80%
3000/mm3
Months Post-Initial Φ Treatment
b
Hip Aspirate
NUCLEATED CELL COUNT
NEUTROPHIL %
Cells /mm3
Neutrophils (%)
80%
3000/mm3
Months Post-Initial Φ Treatment
c
Inflammatory Serology
ESR
CRP
ESR mm/h
CRP mg/dL
30 mm/h
1 mg/dL
+ EFFUSION
− EFFUSION
Months Post-Initial Φ Treatment
d
Liver Function Test
AST
ALT
AST (U/L)
ALT (U/L)
40 U/L
Months Post-Initial Φ Treatment



Phage TEM image
a
b
BF
Activity: Growth Inhibition
c
d
PASA16
Φ83
Planktonic (Hours)
S1
S2
S8
S22
S23
Serum
Synovial Fluid
Activity: Antibodies
e
f
g
h
PASA16
PFU/mL
Φ83
SM+
6
9
12
24
Months
*
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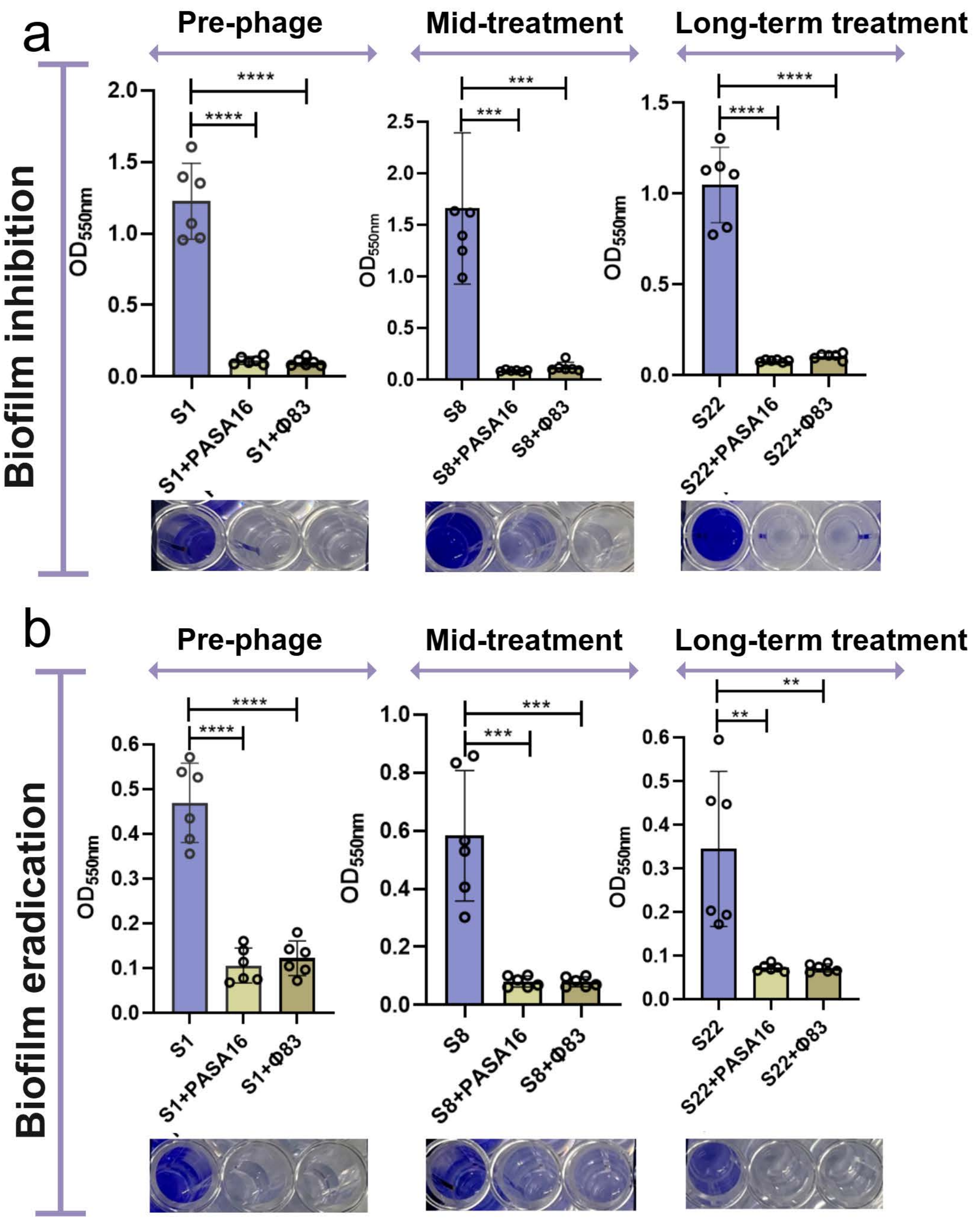
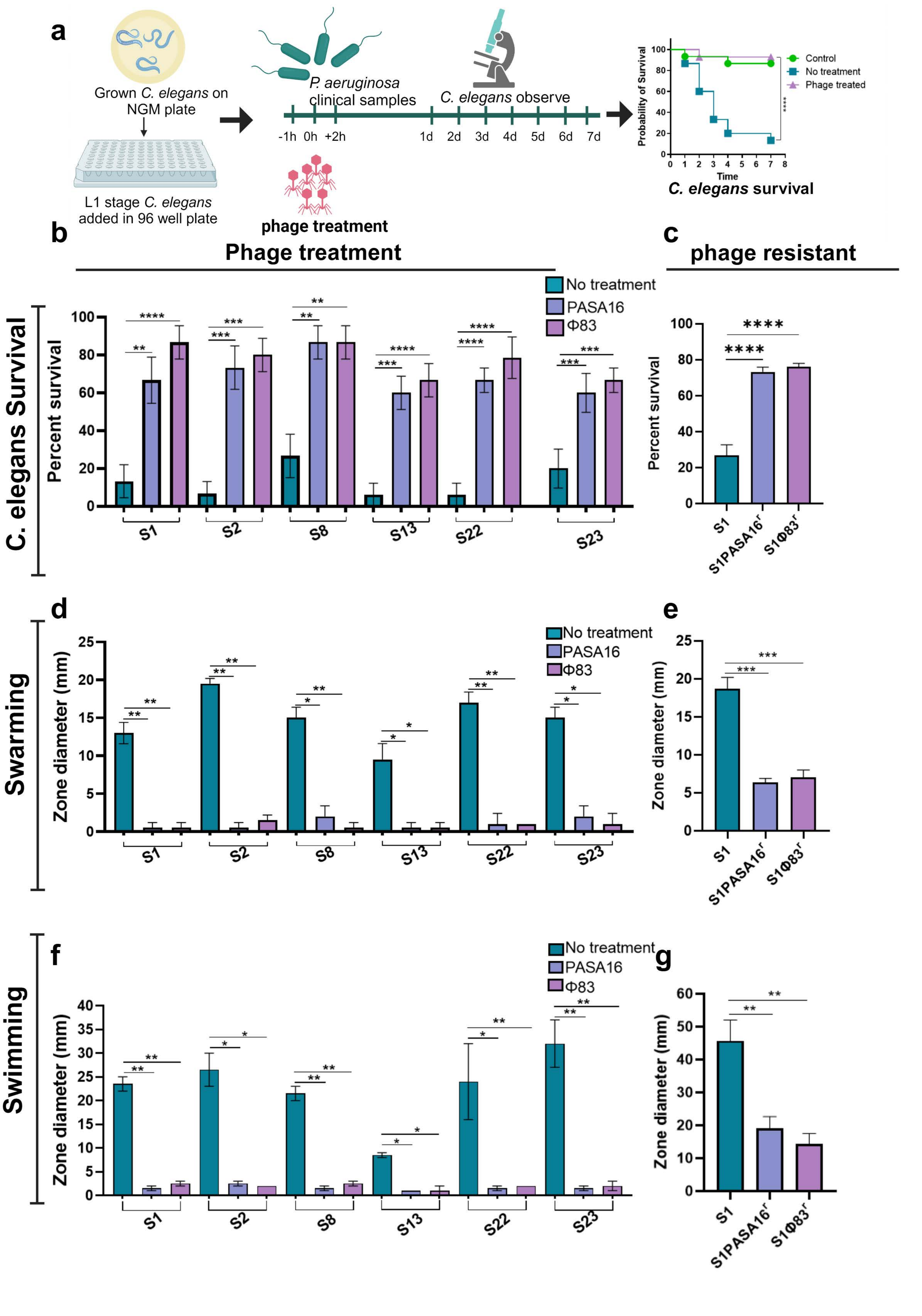

a
Biofilm inhibition
Pre-phage
Mid-treatment
Long-term treatment
OD550nm
S1
S1+PASA16
S1+Φ83
S8
S8+PASA16
S8+Φ83
S22
S22+PASA16
S22+Φ83
b
Biofilm eradication
Pre-phage
Mid-treatment
Long-term treatment
OD550nm
S1
S1+PASA16
S1+Φ83
S8
S8+PASA16
S8+Φ83
S22
S22+PASA16
S22+Φ83


a
Grown C. elegans on NGM plate
L1 stage C. elegans added in 96 well plate
P. aeruginosa clinical samples
C. elegans observe
-1h 0h +2h 1d 2d 3d 4d 5d 6d 7d
phage treatment
Probability of Survival
Time
Control
No treatment
Phage treated
C. elegans survival
b
Phage treatment
C. elegans Survival
Percent survival
No treatment
PASA16
Φ83
S1 S2 S8 S13 S22 S23
c
phage resistant
Percent survival
S1 S1PASA16r S1Φ83r
d
Swarming
Zone diameter (mm)
No treatment
PASA16
Φ83
S1 S2 S8 S13 S22 S23
e
Zone diameter (mm)
S1 S1PASA16r S1Φ83r
f
Swimming
Zone diameter (mm)
No treatment
PASA16
Φ83
S1 S2 S8 S13 S22 S23
g
Zone diameter (mm)
S1 S1PASA16r S1Φ83r